

MIAE 221: Materials Science for Engineers

Chapters 14 & 15: Polymer Structures and Properties (Expanded & Condensed Guide)

Concordia University · Gina Cody School of Engineering

Textbook:

Materials Science and Engineering: An Introduction

(10th Ed., Callister & Rethwisch)

Instructor: Dr. Mamoun Medraj, P.Eng | Curriculum Alignment: Concordia University

(Syllabus Week 10 · Sections 14.1–14.15 & 15.1–15.14)

1. 🎯 30-Second Core Intuition & Plain-English Concept

Polymers are gigantic macromolecular chains made of repeating chemical units (monomers) linked by strong covalent bonds.

The mechanical behavior depends entirely on whether chains are free to slide past one another (**Thermoplastics**) or permanently locked together into an un-meltable 3D net (**Thermosets**).

Key Real-World Distinction

:

- **Thermoplastic** (e.g., Polyethylene, Nylon): Like a bowl of cooked spaghetti. When heated, chains slide freely and the plastic melts. Can be reheated and recycled indefinitely.
- **Thermoset** (e.g., Epoxy, Vulcanized Rubber): Like a baked cake. Chemical cross-links form permanent covalent bridges across all chains. Reheating will not melt it; it will simply char and burn. Cannot be recycled by melting.

2. ⚙️ High-Yield Mathematical Engine & Molecular Metrics

1. Molecular Weight and Degree of Polymerization

Polymer chains have varying lengths, described by statistical distributions:

- **Number-Average Molecular Weight:** $\bar{M}_n = \sum x_i M_i$ (x_i : number fraction).
- **Weight-Average Molecular Weight:** $\bar{M}_w = \sum w_i M_i$ (w_i : weight fraction).
- **Polydispersity Index (PDI):** $PDI = \frac{\bar{M}_w}{\bar{M}_n} \geq 1.0$ (Measures breadth of chain length distribution).
- **Degree of Polymerization (DP):**

$$DP = \frac{\bar{M}_n}{m}$$

where m is the molecular weight of the single repeating monomer unit.

2. Molecular Architectures & Tacticity

- **Architectures:** Linear $\rightarrow$ Branched $\rightarrow$ Cross-linked $\rightarrow$ Network.

- **Tacticity (Stereoisomerism):**

- **Isotactic:** All side R -groups positioned on the
same side
of the chain (crystallizes easily).

- **Syndiotactic:** R -groups alternate regularly from
side to side

- **Atactic:** R -groups positioned
randomly
(prevents chain packing; completely amorphous).

3. Thermal Transitions

- **Glass Transition Temperature (T_g):** Temperature below which amorphous polymer chains freeze into a brittle, glassy state; above T_g , chains become mobile, leathery, and rubbery.
- **Melting Temperature (T_m):** Temperature where crystalline ordered lamellae melt into a disordered liquid.

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

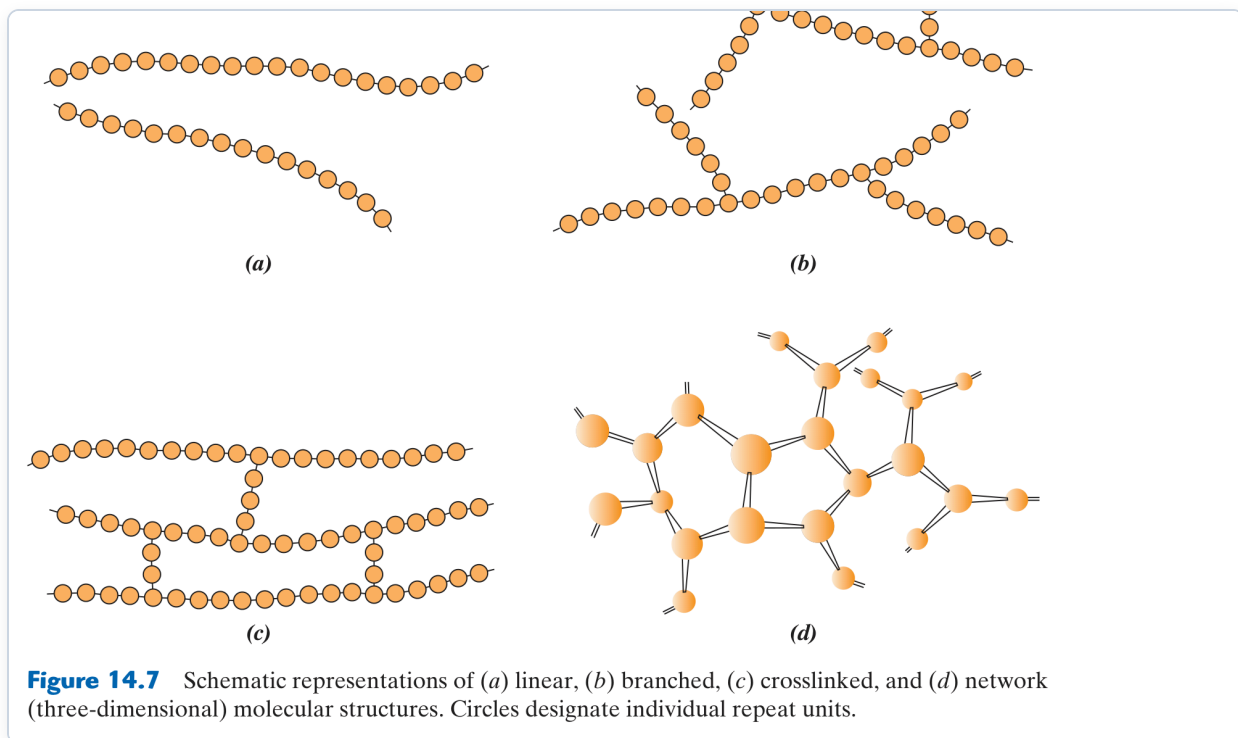


Figure 14.7: Schematic representation of polymer chain structures: (a) Linear, (b) Branched, (c) Crosslinked, and (d) Network.

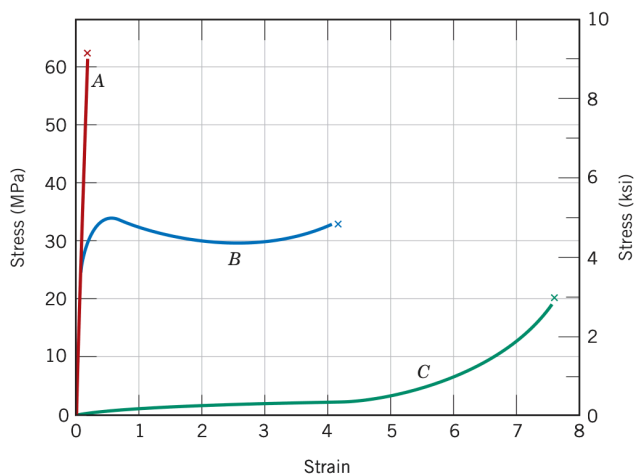


Figure 15.1 The stress-strain behavior for brittle (curve A), plastic (curve B), and highly elastic (elastomeric) (curve C) polymers.

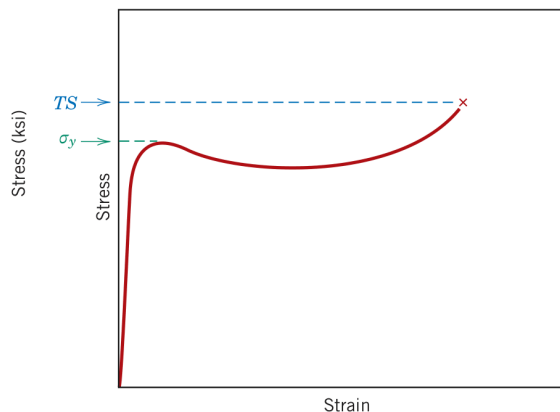


Figure 15.2 Schematic stress-strain curve for a plastic polymer showing how yield and tensile strengths are determined.

Figure 15.1: Stress-strain behavior for polymers: Curve A (Brittle polymer below T_g), Curve B (Plastic polymer with necking/drawing), and Curve C (Highly elastic elastomer).

In-Depth Visual Breakdown:

- **Chain Architectures (Figure 14.7):**

- (a) *Linear*
: High packing density (HDPE); high crystallinity and strength.
- (b) *Branched*
: Side branches prevent tight packing (LDPE); lower density and greater flexibility.
- (c) *Crosslinked*
: Adjacent chains joined by covalent bonds (vulcanized rubber with sulfur crosslinks); imparts elasticity.
- (d) *Network*
: Fully 3D covalent crosslinking (epoxies, phenolics); rigid and completely non-meltable.

- **Polymer Stress-Strain Curves (Figure 15.1):**

- Unlike metals where necking leads directly to failure, semicrystalline polymers (Curve B) experience **drawing**: the polymer necks, and then the neck propagates down the entire gage length as disordered polymer chains align parallel to the tensile axis, dramatically strengthening the necked region!

4. 🏴‍☠️ Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**

- Calculating repeat unit molecular weight m and finding DP from given $\bar{M}_n$.
- Comparing properties above and below T_g (e.g., rubber car tires shattering like glass in liquid nitrogen).
- Distinguishing condensation (step-growth, releases water by-product) vs. addition (chain-growth) polymerization.

- **Concordia Exam Traps:**

- **Repeat Unit Mass (m) Calculation:** Forgetting to account for double bonds opening up. Polyethylene monomer is ethylene (C_2H_4), so $m = 2(12.011) + 4(1.008) = 28.05 \text{ g/mol}$. In vinyl chloride ($\text{C}_2\text{H}_3\text{Cl}$), you replace one H with Cl (35.45 g/mol).

5. 💡 Master Exam Problem & Step-by-Step Solution Framework

Problem:

A polyvinyl chloride (PVC) pipe material has a number-average molecular weight $\bar{M}_n = 125,000 \text{ g/mol}$. (a) Calculate the repeat unit molecular weight m . (b) Determine its degree of polymerization DP .

- **Step 1: Determine Chemical Formula of PVC Repeat Unit**

PVC repeat unit is $[-\text{CH}_2 - \text{CHCl}-]_n$ (derived from $\text{C}_2\text{H}_3\text{Cl}$)

:

$$\text{Carbon: } 2 \times 12.011 = 24.022 \text{ g/mol}$$

$$\text{Hydrogen: } 3 \times 1.008 = 3.024 \text{ g/mol}$$

$$\text{Chlorine: } 1 \times 35.45 = 35.450 \text{ g/mol}$$

$$m = 24.022 + 3.024 + 35.450 = 62.496 \text{ g/mol}$$

- **Step 2: Calculate Degree of Polymerization DP**

$$DP = \frac{\bar{M}_n}{m} = \frac{125,000 \text{ g/mol}}{62.496 \text{ g/mol}} = 2,000.1 \approx 2,000 \text{ repeat units}$$
